

NETRA-MAIL

Multilingual phishing intelligence.
Investigation-ready email forensics.

Problem Statement ID: 26106 — confirm

PS Title: [Exact portal title to be confirmed]

Theme: [Confirm portal theme]

PS Category: Software

Team ID: [To be added]

Team Name: [Registered name to be added]

DETECT

Multilingual text, URLs, QR and attachments

INVESTIGATE

Message, URL and relay evidence

PRESERVE

Case timeline and forensic report

Challenges and problems

Language barriers in phishing detection

Regional scripts, transliteration and mixed-language messages need contextual analysis.

Deception across multiple email surfaces

Spoofed senders, credential links, payment requests and image-based lures obscure intent.

Fragmented investigation and missing context

A risk label alone does not explain relay paths, related indicators or evidence history.

Proposed solution — detailed explanation

1. Multilingual phishing intelligence

Combine language/script cues with urgency, OTP, credential and financial-fraud indicators.

2. Multi-layer email inspection

Fuse text/ML findings, sender alignment, domain/URL checks, static attachments and QR/OCR signals.

3. Investigation and evidence lifecycle

Open the exact email in the SOC dashboard; inspect findings, correlate cases and export reports.

Language profiles: English · Hindi · Tamil · Telugu · Kannada · Malayalam. Per-language accuracy remains to be measured.

Innovation and uniqueness

MULTILINGUAL

English + 5 Indian
language profiles

EXPLAINABLE

Risk score + reasons
+ evidence limits

FORENSIC

SHA-256 + custody
+ encrypted evidence

CONNECTED

Gmail badge → exact
SOC investigation

How it addresses the problem

Language-aware review

Native scripts + Romanized cues; language is context, never a malicious verdict.

From alert to evidence

Relay/IP context and related indicators help analysts investigate recurring campaigns.

Controlled and transparent analysis

Manual scan, scoped access and explicit labels when original headers are unavailable.

Methodology and process

01 Acquire & authenticate

User selects Gmail content or submits EML; API checks identity, origin and request limits.

02 Normalize & extract

Parse MIME, sender fields, relay headers, URLs and static attachments; mask sensitive fields.

03 Analyze multilingual signals

Detect script / Romanized cues; inspect urgency, credentials, financial fraud and mixed scripts.

04 Fuse evidence & explain

Combine ML, rules, URL/domain checks and authentication evidence with confidence and limitations.

05 Investigate & preserve

Link indicators and cases, store encrypted evidence, record custody and generate forensic reports.

Technology stack — mapped to the prototype



Python
Analysis engine



FastAPI
REST + validation



JavaScript
Chrome MV3



scikit-learn
Risk models



Streamlit
SOC interface



SQLite
Forensic ledger



OpenCV
QR decoding



RapidOCR
Image text



dkimpy / DNS
Authentication



cryptography
AES-GCM



Docker
Isolated worker*



Render
Hosted services

*Isolated inspection / reputation require separate configuration. Missing original bytes or provider data are labelled, not assumed safe.

System architecture and working flow

INPUT LAYER

Gmail extension
Original EML / MIME
Scoped API key



FASTAPI GATEWAY

Origin + identity validation
Size limits · ingestion · orchestration

SOURCES

Received headers
DNS + public IPs
Original bytes
Optional reputation*



MULTI-LAYER ANALYSIS

Multilingual NLP + ML scoring
Sender / DKIM / DMARC evidence
URL + domain + IP intelligence
Attachment / QR / OCR signals

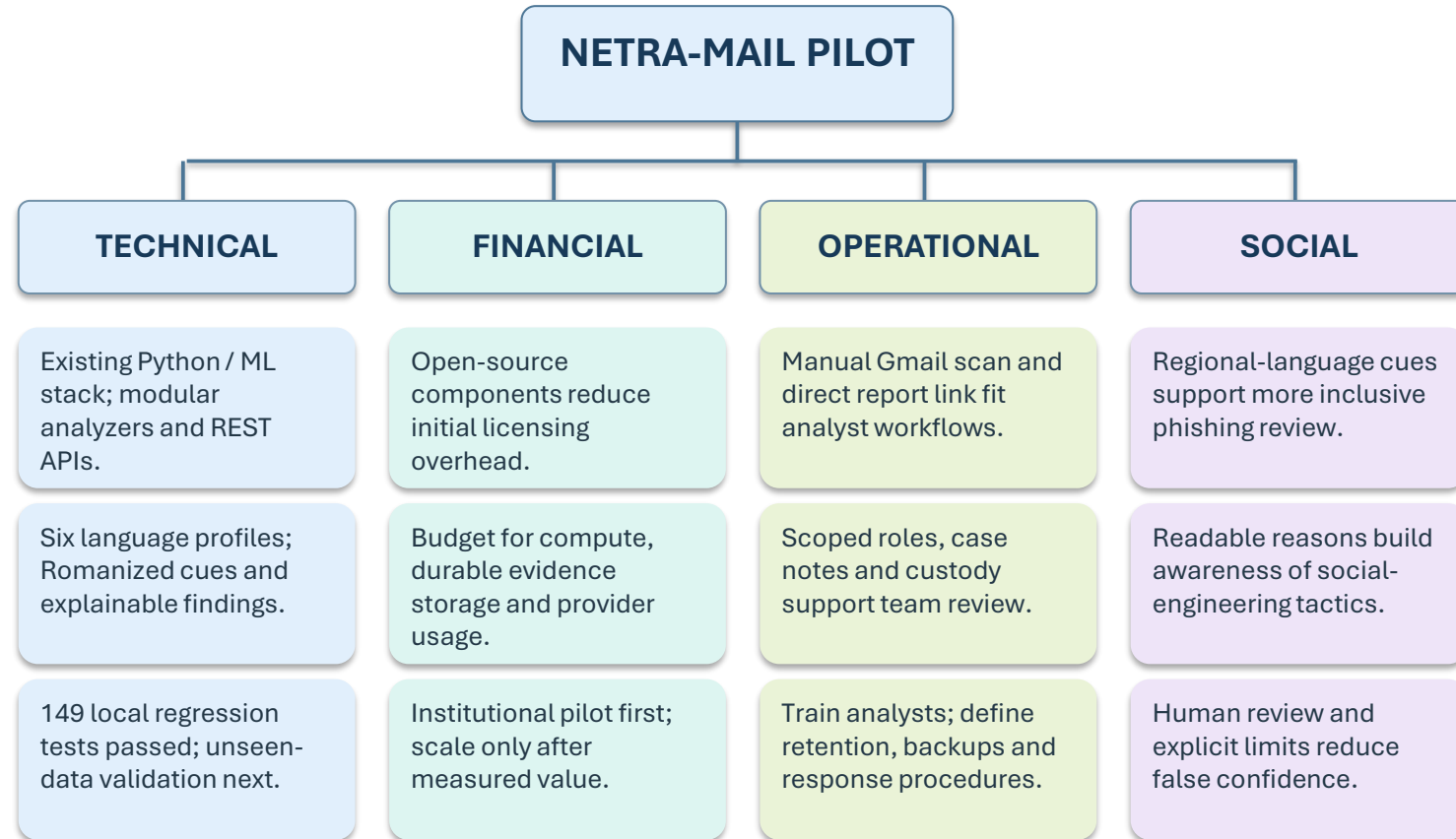
PRESERVATION

SQLite + SHA-256
AES-GCM evidence
Versioned records

SOC INVESTIGATION

Findings · relay trace · graph
Cases · custody · PDF/HTML reports

Analysis of feasibility and viability



Risks and mitigation strategies

TECHNICAL CHALLENGES

- Missing originals → EML / authorized MIME fetch.
- Language/model errors → labelled per-language tests.
- Provider outages → timeouts and unavailable labels.

FINANCIAL CHALLENGES

- Hosting/storage costs → staged institutional pilot.
- Paid intelligence → optional, budgeted providers.
- Data growth → retention and backup policy.

OPERATIONAL CHALLENGES

- Public access → dashboard login before real use.
- False alerts → analyst review and feedback.
- Evidence continuity → durable disk + restore drills.

Pilot readiness is distinct from production readiness. Regression tests do not establish phishing accuracy or legal admissibility.

Potential impact on the target audience

Citizens and institutional email users

Language-aware warnings expose credential theft, payment pressure and deceptive links.

SOC analysts and campus IT teams

One investigation view connects findings, observed infrastructure, campaigns and evidence.

Incident-response and review teams

Versioned records, custody events and forensic reports improve handover and review.

Regional-language communities

English, Hindi, Tamil, Telugu, Kannada and Malayalam cues broaden contextual coverage.

Benefits — social, economic and resource use

- Social: clearer phishing awareness and regional-language inclusion.
- Economic: aim to reduce manual review effort and fraud exposure.
- Resource use: digital reports and user-initiated scans avoid unnecessary processing.

Measure: per-language precision/recall · false-positive rate · median/p95 scan latency · review time · report completeness. No outcome claims yet.

Adoption, sustainability and measurable value

TARGET USERS / EARLY ADOPTERS

University IT cells, institutional SOC's and small organizations handling suspicious email reports.

VALUE-ADDED CAPABILITIES

Multilingual cues · QR/OCR inspection · explainable risk
Authentication evidence · campaign links · traceable reports

SUSTAINABILITY MODEL — PROPOSED

Institution-managed hosting; optional paid intelligence. Evaluate support subscriptions after pilot validation.

PHASED ROLLOUT

Institution
pilot



Measure &
calibrate



Multi-team
adoption

Reference research and standards

[DKIM — RFC 6376](#)

Signature verification over original message bytes; signing-domain evidence.

[SPF — RFC 7208](#)

Sender authorization requires trusted SMTP connection context.

[DMARC — RFC 7489](#)

From-domain alignment using DKIM and trusted SPF evidence.

[Digital forensics — NIST SP 800-86](#)

Collection → examination → analysis → reporting.

Project source and reproducible evidence

github.com/nitheenkumar25bcy37-alt/NETRA-MAIL

Prototype evidence and validation roadmap

IMPLEMENTED PROTOTYPE

Chrome MV3 + FastAPI + Streamlit; multilingual rules; QR/OCR integration; authentication evidence; cases, correlation, encrypted storage and report generation.

MULTILINGUAL DESIGN BASIS

Unicode script detection + contextual security cues.
Native scripts and Romanized/transliterated patterns.
Output: language, method, cues, confidence and limits.

EVALUATION TO COMPLETE

- Unseen benign/phishing corpus with language labels.
- Native + Romanized + mixed-script cases.
- False positives, recall and latency by language.
- Original-MIME and missing-evidence scenarios.
- Hosted access, persistence and recovery checks.

149 local automated tests passed; no claimed accuracy.